

Supplementary Material

Experimental and DFT-Assisted Raman Characterisation of Isolated β -N-Acetylmuramic Acid

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S1. Computational details

Geometry optimisation and harmonic vibrational frequency calculations were performed with Gaussian 16 using the route section

```
# opt=(calcfc,tight) freq=raman b3lyp/6-311++g(d,p)
empiricaldispersion=gd3bj geom=connectivity int=ultrafine scf=(tight,xqc)
```

on the β anomer of N-acetylmuramic acid taken from the PubChem 3D conformer record (CID 5462244). The optimisation converged to a stationary point and the calculation reached normal termination. The optimised structure comprises 39 atoms ($\text{C}_{11}\text{H}_{19}\text{NO}_8$) and gave 111 normal modes, none imaginary. Harmonic wavenumbers were scaled by 0.980 below 1800 cm^{-1} and 0.967 above.

S2. Experimental dataset

98 independent Raman spectra of N-acetylmuramic acid powder were acquired on a B&W Tek BWS465-785H system (784.92 nm, 2048-pixel detector, two detector readings averaged per spectrum) across five integration times (5, 10, 15, 20, 25 s) and sixteen laser power levels (5–80% in 5% steps). No detector saturation occurred at any condition. The 40 spectra with $\text{SNR} \geq 100$ were used to construct the reference spectrum reported in the main text; their mean pairwise correlation coefficient is 0.957.

S3. Optimised Cartesian coordinates

S4. Complete list of calculated normal modes

Table S2: All 111 calculated normal modes. ν : unscaled harmonic wavenumber; ν_s : scaled wavenumber; S : Raman activity ($\text{\AA}^4/\text{amu}$); A : IR intensity (km/mol).

#	ν	ν_s	S	A	#	ν	ν_s	S	A
1	34.82	34.13	0.878	0.24	57	1113.10	1090.84	4.810	26.71
2	40.87	40.05	1.180	1.29	58	1130.03	1107.43	6.554	236.52
3	42.12	41.27	0.253	0.85	59	1139.36	1116.58	3.055	276.09
4	57.31	56.16	0.221	0.30	60	1147.58	1124.63	2.746	103.06
5	60.11	58.90	0.466	1.02	61	1158.55	1135.38	1.888	35.56
6	71.46	70.03	0.400	2.22	62	1179.81	1156.21	3.271	19.03
7	84.59	82.90	1.031	8.94	63	1196.36	1172.43	4.411	9.66
8	90.70	88.88	0.103	2.52	64	1216.46	1192.13	2.990	0.21
9	107.93	105.77	0.123	3.53	65	1232.88	1208.22	2.125	38.71
10	120.61	118.20	0.119	4.76	66	1254.46	1229.37	0.970	5.11
11	148.73	145.75	0.087	4.91	67	1267.63	1242.27	3.847	11.20

#	ν	ν_s	S	A	#	ν	ν_s	S	A
12	175.55	172.04	0.847	3.11	68	1282.76	1257.10	6.022	125.07
13	182.62	178.97	0.081	3.27	69	1310.62	1284.41	1.484	150.91
14	213.68	209.41	0.203	0.24	70	1313.88	1287.60	4.543	16.90
15	230.14	225.54	0.284	56.07	71	1332.86	1306.21	1.307	4.22
16	241.59	236.76	1.744	50.31	72	1340.36	1313.55	2.719	12.40
17	244.28	239.40	1.135	0.86	73	1351.86	1324.82	0.715	4.98
18	264.14	258.86	1.077	13.63	74	1361.58	1334.35	1.938	2.20
19	285.05	279.35	0.571	23.88	75	1373.15	1345.69	5.360	4.35
20	294.95	289.05	3.245	1.37	76	1384.29	1356.60	2.107	5.58
21	301.70	295.66	2.174	73.20	77	1387.00	1359.26	10.986	10.60
22	314.12	307.84	1.561	12.38	78	1396.66	1368.73	2.703	37.89
23	340.59	333.78	0.949	3.77	79	1400.65	1372.63	2.211	23.11
24	345.33	338.42	1.337	21.34	80	1407.31	1379.17	3.795	4.71
25	385.45	377.74	1.774	8.89	81	1415.92	1387.61	7.768	8.86
26	411.66	403.43	3.917	25.26	82	1436.72	1407.98	5.368	48.93
27	423.89	415.42	1.354	38.83	83	1438.71	1409.93	3.856	9.26
28	449.53	440.54	0.633	81.10	84	1462.43	1433.18	2.650	5.42
29	469.85	460.45	1.027	32.68	85	1473.40	1443.94	5.230	11.66
30	504.26	494.17	2.844	19.54	86	1489.54	1459.75	7.552	15.05
31	523.85	513.37	2.287	10.96	87	1489.73	1459.94	5.450	3.78
32	566.96	555.62	3.176	3.86	88	1500.82	1470.80	6.370	1.81
33	581.59	569.96	2.284	9.63	89	1501.81	1471.77	4.275	7.42
34	584.97	573.27	1.838	93.18	90	1566.57	1535.24	2.154	257.10
35	594.02	582.14	0.910	55.28	91	1757.98	1722.82	12.805	308.61
36	620.81	608.39	0.625	18.43	92	1795.14	1759.24	10.169	328.18
37	632.03	619.39	0.459	11.06	93	2934.52	2837.68	48.614	47.10
38	639.48	626.69	0.830	24.68	94	2977.52	2879.27	139.250	7.80
39	653.21	640.15	1.868	57.36	95	2990.04	2891.37	11.131	51.84
40	706.31	692.19	1.470	36.92	96	3001.66	2902.61	83.689	16.66
41	746.20	731.27	5.290	26.55	97	3033.60	2933.49	24.378	35.48
42	809.38	793.19	7.406	9.66	98	3042.37	2941.98	181.917	7.57
43	890.78	872.97	3.213	22.88	99	3047.82	2947.24	190.231	15.33
44	923.28	904.81	2.333	3.01	100	3052.20	2951.48	22.588	11.52
45	945.58	926.67	3.568	6.15	101	3076.75	2975.22	31.609	12.31
46	999.46	979.47	2.644	8.69	102	3091.05	2989.04	72.808	23.83
47	1020.26	999.85	2.380	10.16	103	3108.35	3005.77	59.045	11.25
48	1026.75	1006.22	1.967	92.29	104	3122.14	3019.11	49.657	12.08
49	1042.37	1021.52	2.308	10.64	105	3129.29	3026.02	58.738	10.63
50	1049.19	1028.21	0.890	118.98	106	3135.97	3032.48	65.276	10.69
51	1056.02	1034.90	0.202	23.67	107	3565.54	3447.87	84.951	176.05
52	1063.47	1042.20	7.653	72.50	108	3752.61	3628.77	147.673	80.33
53	1076.34	1054.82	1.783	39.46	109	3804.46	3678.91	46.110	39.45
54	1089.60	1067.80	0.588	205.57	110	3830.63	3704.22	146.915	47.43
55	1094.55	1072.66	12.446	5.23	111	3853.40	3726.23	113.263	45.54
56	1101.87	1079.83	0.618	32.85					

S5. Data files

The following are provided alongside this document:

- `Table_DFT_all_modes.csv` — all 111 modes with frequency, scaled frequency, Raman activity,

Table S1: Optimised Cartesian coordinates of β -N-acetylmuramic acid (B3LYP/6-311+G(d,p), Å).

#	El.	<i>x</i>	<i>y</i>	<i>z</i>	#	El.	<i>x</i>	<i>y</i>	<i>z</i>
1	O	-2.680794	0.670787	-0.094904	21	H	0.115079	-0.742934	-0.965928
2	O	1.069463	-0.759373	0.872185	22	H	-0.461724	1.360374	1.192032
3	O	-1.067922	-2.615191	0.317311	23	H	-1.448390	-1.028364	1.618181
4	O	-1.840721	2.758593	-0.464037	24	H	-2.405640	-1.001894	-1.294143
5	O	-4.955878	-0.860424	-0.465147	25	H	-1.460230	1.142196	-1.687055
6	O	0.942881	3.225032	1.408684	26	H	-3.949603	-1.118455	1.337915
7	O	3.357269	-2.528753	-1.283455	27	H	-3.830150	-2.425509	0.133294
8	O	2.634316	-0.412110	-1.416760	28	H	1.474733	1.235877	-0.949221
9	N	0.859442	1.760669	-0.340525	29	H	1.285482	-2.711805	0.221839
10	C	-0.072282	-0.460237	0.078650	30	H	-1.622947	-3.135841	0.904684
11	C	-0.301541	1.055014	0.154576	31	H	-2.582451	2.996868	-1.030119
12	C	-1.307206	-1.222243	0.548677	32	H	3.458064	-1.245169	1.811664
13	C	-2.537918	-0.741338	-0.233338	33	H	3.527990	-2.951307	1.307290
14	C	-1.575401	1.409778	-0.622512	34	H	2.293590	-2.407382	2.465753
15	C	-3.843175	-1.342297	0.266764	35	H	-4.912686	0.102812	-0.432622
16	C	1.889137	-1.825745	0.442405	36	H	3.455207	3.418734	0.666429
17	C	1.444020	2.757941	0.401236	37	H	3.199299	2.597780	-0.898148
18	C	2.863583	-2.130825	1.578298	38	H	2.581649	4.240235	-0.627046
19	C	2.646027	-1.473067	-0.837832	39	H	3.815808	-2.253317	-2.092008
20	C	2.758411	3.268675	-0.158613					

IR intensity, reduced mass, atom-fraction descriptors and proposed assignment.

- `Table_DFT_fingerprint_region.csv` — the 67 modes lying within 400–1800 cm^{-1} .
- `DFT_simulated_spectrum.csv` — Lorentzian-broadened simulated spectrum on a 0.5 cm^{-1} grid.

[Add the processed experimental mean spectrum as a CSV, and consider depositing raw spectra plus the Gaussian log in Zenodo before submission.]